SPC315 – Thin Walled Structures
SPC316 – Aerospace Vehicles Structural Analysis & Design

Wing Design for Utility Category General Aviation Aircraft

Spring 2026

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Abstract

This report presents the preliminary aerodynamic and structural design of a wing for a utility-category general aviation aircraft. The work begins with the airfoil selection process based on the required cruise and stall lift coefficients. The selected airfoil is then used to define the main wing geometric parameters, including aspect ratio, taper ratio, span, root chord, tip chord, dihedral angle, setting angle, and washout. Finally, the spanwise lift distribution is evaluated using Prandtl's Lifting-Line Theory implemented in MATLAB. The results confirm that the proposed wing satisfies the required cruise lift coefficient and provides a sufficient take-off lift margin.

Contents

Abstract **i**

1 Introduction **1**
- 1.1 Project Overview . . . 1
- 1.2 Given Aircraft Requirements . . . 1
- 1.3 Flight Conditions Used in the Design . . . 1
- 1.4 Meaning of the Main Aerodynamic Parameters . . . 2
- 1.5 Design Methodology . . . 2

2 Airfoil Selection and XFLR5 Verification **4**
- 2.1 Purpose of the Airfoil Study . . . 4
- 2.2 Required Lift Coefficient Calculations . . . 4
  - 2.2.1 Aircraft Ideal Cruise Lift Coefficient . . . 4
  - 2.2.2 Wing Cruise Lift Coefficient . . . 5
  - 2.2.3 Airfoil Ideal Lift Coefficient . . . 5
  - 2.2.4 Aircraft Maximum Lift Coefficient . . . 5
  - 2.2.5 Wing Maximum Lift Coefficient . . . 6
  - 2.2.6 Required Airfoil Maximum Lift Coefficient . . . 6
- 2.3 Initial Research-Based Airfoil Selection . . . 6
- 2.4 Initial Selection Matrix . . . 7
- 2.5 NACA 2412 Design Calculations . . . 7
- 2.6 XFLR5 Analysis of NACA 23015 . . . 8
- 2.7 XFLR5 Comparison Between Candidate Airfoils . . . 10
- 2.8 Comparison Between NACA 2412 and NACA 23015 . . . 12
- 2.9 Final Airfoil Decision for CAD Modelling . . . 13

3 Wing Geometric Design **14**

3.1 Wing Planform Selection . . . 14
3.2 Wing Setting Angle . . . 14
3.3 Aspect Ratio and Wingspan . . . 14
3.4 Taper Ratio and Chord Dimensions . . . 14
3.5 Dihedral Angle . . . 15
3.6 Wing Twist . . . 15
3.7 Wing Geometry Summary . . . 15

4 Mechanical Wing Design Using NACA 23015 **16**
4.1 Design Overview . . . 16
4.2 Main I-Beam Spar Design . . . 16
4.2.1 Main Spar at the Root . . . 17
4.2.2 Main Spar at the Tip . . . 17
4.3 Rear I-Beam Spar Design . . . 18
4.3.1 Rear Spar at the Root . . . 18
4.3.2 Rear Spar at the Tip . . . 19
4.4 C-Shaped Stringer Design . . . 20
4.4.1 C-Stringers at the Root . . . 20
4.4.2 C-Stringers at the Tip . . . 21
4.5 Rib Design and Chord Distribution . . . 22
4.6 Wing Skin Design . . . 24
4.7 Complete CAD Assembly . . . 24
4.8 Mechanical Design Summary . . . 26

5 Spanwise Lift Distribution **27**
5.1 Theoretical Background . . . 27
5.2 MATLAB Implementation . . . 27
5.3 Solver Inputs . . . 29
5.4 Spanwise Lift Distribution Results . . . 29
5.5 Total Wing Lift Coefficient Verification . . . 29
5.6 Estimated Wing Lift at Take-Off . . . 30

6 Conclusion **31**

List of Tables

1.1 Group 7 aircraft design requirements . . . 1
1.2 Aerodynamic flight-condition inputs used for airfoil analysis and XFLR5 verification . . . 2

2.1 Aircraft data used for the airfoil selection calculations . . . 4
2.2 Summary of required lift coefficient calculations . . . 6
2.3 Initial candidate airfoil comparison . . . 6
2.4 Initial airfoil selection matrix . . . 7
2.5 NACA 2412 properties used in the initial design . . . 7
2.6 XFLR5 setup and extracted results for NACA 23015 . . . 9
2.7 Comparison between NACA 2412 and NACA 23015 . . . 13

3.1 Final wing geometry parameters . . . 15

4.1 Main I-beam spar characteristics at the wing root . . . 17
4.2 Main I-beam spar characteristics at the wing tip . . . 17
4.3 Rear I-beam spar characteristics at the wing root . . . 18
4.4 Rear I-beam spar characteristics at the wing tip . . . 19
4.5 C-shaped stringer characteristics at the wing root . . . 20
4.6 C-shaped stringer characteristics at the wing tip . . . 21
4.7 Rib station locations and local chord lengths for the NACA 23015 CAD model . . . 23
4.8 Wing skin design characteristics . . . 24

5.1 Inputs used in the MATLAB lifting-line solver . . . 29
5.2 Spanwise lift coefficient distribution from MATLAB . . . 29

List of Figures

2.1 XFLR5 pressure coefficient distribution and boundary-layer visualization for NACA 23015 at $\alpha = 6°$ 9
2.2 XFLR5 aerodynamic distributions for NACA 23015 showing moment coefficient, total angle, lift coefficient, local lift, and total drag behavior 10
2.3 XFLR5 take-off comparison between candidate airfoils 11
2.4 XFLR5 cruise comparison between candidate airfoils 12

4.1 Main I-beam spar cross-section at the wing root 17
4.2 Main I-beam spar cross-section at the wing tip 18
4.3 Rear I-beam spar cross-section at the wing root 19
4.4 Rear I-beam spar cross-section at the wing tip 20
4.5 C-shaped stringer cross-section at the wing root 21
4.6 C-shaped stringer cross-section at the wing tip 21
4.7 Isometric CAD view of the spars and C-shaped stringers 22
4.8 Isometric CAD view of the ribs and their spanwise positions 23
4.9 Isometric CAD view of the wing skin 24
4.10 Full internal CAD design showing spars, stringers, ribs, and skin 25
4.11 Complete isometric CAD view of the designed NACA 23015 wing 26

5.1 MATLAB implementation of the lifting-line solver 28

Chapter 1

Introduction

1.1 Project Overview

The objective of this project is to design a wing for a utility-category general aviation aircraft. The design process includes airfoil selection, wing planform sizing, aerodynamic analysis, CAD modelling, and preparation for the mechanical structural design phase. The selected wing must satisfy the given aircraft performance requirements while maintaining acceptable aerodynamic efficiency, stability, manufacturability, and structural feasibility.

The design was carried out in two main levels. First, the aerodynamic design was developed based on the given aircraft requirements, including wing area, maximum take-off mass, cruise speed, and stall speed. These values were used to calculate the required lift coefficients at cruise and take-off conditions. Second, the airfoil and wing geometry were checked using XFLR5 to compare the aerodynamic behavior of different airfoils before selecting the final airfoil used in the CAD model.

1.2 Given Aircraft Requirements

Parameter	Value
Wing area, S	17.1 m^2
Maximum take-off mass, m_{TO}	2,717 kg
Aircraft weight, $W = m_{TO}g$	26,664 N
Cruise speed, V_c	233 knots at 25,000 ft
Stall speed, V_s	100 knots at sea level

Table 1.1: Group 7 aircraft design requirements

1.3 Flight Conditions Used in the Design

Two flight conditions were considered in the aerodynamic design: cruise and take-off. The cruise condition represents the normal high-speed operating point of the aircraft, while the take-off condition represents the high-lift low-speed case. These two cases are important because they define the minimum aerodynamic requirements that the selected airfoil and wing must satisfy.

The aerodynamic inputs used for both flight conditions are summarized in Table 1.2. These values were later used in the airfoil selection and XFLR5 analysis.

Flight Condition	Density ρ (kg/m^3)	Dynamic Viscosity μ ($Pa \cdot s$)	Kinematic Viscosity ν (m^2/s)	Reynolds Number Re	Mach	Wing C_L	Airfoil C_l	$Re\sqrt{C_l}$	$M\sqrt{C_l}$
Cruise — 233 knots at 25,000 ft	0.549	1.53×10^{-5}	2.79×10^{-5}	6.7×10^6	0.39	0.415	0.461	4.55×10^6	0.265
Take-off — 100 knots at sea level	1.225	1.789×10^{-5}	1.46×10^{-5}	5.5×10^6	0.15	1.515	1.683	7.13×10^6	0.195

Table 1.2: Aerodynamic flight-condition inputs used for airfoil analysis and XFLR5 verification

1.4 Meaning of the Main Aerodynamic Parameters

The density ρ represents the air density at the selected altitude. At cruise altitude, the density is lower than sea level, which reduces the available aerodynamic lift for the same speed and wing area. Therefore, the cruise lift coefficient must be calculated using the air density at 25,000 ft.

The dynamic viscosity μ and kinematic viscosity ν are used to define the Reynolds number. The Reynolds number is important because it affects the airfoil boundary-layer behavior, transition, separation, drag, and stall characteristics. In this project, the cruise Reynolds number is approximately 6.7×10^6, while the take-off Reynolds number is approximately 5.5×10^6.

The Mach number indicates the compressibility level of the flow. Since both Mach numbers are below 0.4, the aircraft operates in the low-subsonic range. Therefore, a straight unswept wing is suitable, and sweep is not required for drag-divergence considerations.

The wing lift coefficient C_L represents the lift coefficient required from the complete finite wing. The airfoil lift coefficient C_l is slightly higher than the wing lift coefficient because it represents the two-dimensional airfoil section requirement before finite-wing losses and corrections are included.

1.5 Design Methodology

The report is divided into four main technical stages. First, the required lift coefficients are calculated from the aircraft weight, wing area, cruise speed, stall speed, and atmospheric conditions. Second, candidate airfoils are compared using aerodynamic and handling criteria. Third, XFLR5 is used to compare the candidate airfoils under cruise and take-off conditions. Finally, the selected airfoil is used to define the wing geometry and prepare the CAD mechanical design.

The general design workflow is:

1. Define the given aircraft requirements.
2. Calculate the required cruise and take-off lift coefficients.
3. Determine the Reynolds number and Mach number for the operating conditions.
4. Compare the candidate airfoils using research-based data.
5. Verify the airfoil behavior using XFLR5.
6. Select the final airfoil for CAD modelling.

7. Define the wing geometry, including span, chords, taper ratio, dihedral, and twist.
8. Develop the internal wing structure using spars, ribs, stringers, and skin.

This methodology ensures that the final wing design is based on both analytical calculations and aerodynamic software verification.

Chapter 2

Airfoil Selection and XFLR5 Verification

2.1 Purpose of the Airfoil Study

The airfoil selection process was performed in two stages. First, a research-based selection was carried out using the aircraft design requirements and airfoil selection criteria. Based on this stage, NACA 2412 was found to be a suitable airfoil for the required general aviation utility aircraft. Therefore, the first CAD wing model was designed using NACA 2412.

After that, further aerodynamic investigation was performed using XFLR5 software. Several candidate airfoils were compared under take-off and cruise conditions. From the XFLR5 results, NACA 23015 showed strong aerodynamic performance and also provided a larger thickness ratio of 15%, which gives more internal space for spars, ribs, stringers, and skin. Therefore, NACA 23015 was selected for the mechanical CAD wing design phase.

2.2 Required Lift Coefficient Calculations

The aircraft design requirements used in the airfoil selection process are shown in Table 2.1.

Parameter	Value
Wing area, S	17.1 m^2
Maximum take-off mass, m_{TO}	2717 kg
Aircraft weight, $W = m_{TO}g$	26664 N
Cruise speed, V_c	233 knots = 119.86 m/s
Stall speed, V_s	100 knots = 51.44 m/s
Air density at 25,000 ft, ρ_{25}	0.5497 kg/m^3
Sea-level air density, ρ_{SL}	1.225 kg/m^3

Table 2.1: Aircraft data used for the airfoil selection calculations

2.2.1 Aircraft Ideal Cruise Lift Coefficient

The aircraft cruise lift coefficient is calculated from the steady-level flight lift equation:

$$L = W = \frac{1}{2}\rho V^2 S C_L \tag{2.1}$$

Therefore:

$$C_{L_C} = \frac{2W}{\rho V_c^2 S} \tag{2.2}$$

Substituting the cruise data:

$$C_{L_C} = \frac{2(2717)(9.81)}{(0.5497)(119.86)^2(17.1)} \tag{2.3}$$

$$C_{L_C} = 0.394 \tag{2.4}$$

This means that the aircraft must operate at approximately $C_L = 0.394$ during cruise.

2.2.2 Wing Cruise Lift Coefficient

A correction factor of 0.95 is used to account for the aircraft body contribution. The required wing cruise lift coefficient is:

$$C_{L_{C_w}} = \frac{C_{L_C}}{0.95} \tag{2.5}$$

$$C_{L_{C_w}} = \frac{0.394}{0.95} = 0.415 \tag{2.6}$$

2.2.3 Airfoil Ideal Lift Coefficient

The airfoil ideal lift coefficient is estimated from the wing cruise lift coefficient using the correction factor 0.9:

$$C_{l_i} = \frac{C_{L_{C_w}}}{0.9} \tag{2.7}$$

$$C_{l_i} = \frac{0.415}{0.9} = 0.461 \tag{2.8}$$

Therefore, the selected airfoil should have good aerodynamic efficiency near $C_l = 0.461$.

2.2.4 Aircraft Maximum Lift Coefficient

The maximum aircraft lift coefficient is calculated from the stall speed condition at sea level:

$$C_{L_{\max}} = \frac{2W}{\rho_{SL} V_s^2 S} \tag{2.9}$$

$$C_{L_{\max}} = \frac{2(2717)(9.81)}{(1.225)(51.44)^2(17.1)} \tag{2.10}$$

$$C_{L_{\max}} = 1.44 \tag{2.11}$$

2.2.5 Wing Maximum Lift Coefficient

Using the same body correction factor:

$$C_{L_{\max -w}} = \frac{C_{L_{\max}}}{0.95} \tag{2.12}$$

$$C_{L_{\max -w}} = \frac{1.44}{0.95} = 1.515 \tag{2.13}$$

2.2.6 Required Airfoil Maximum Lift Coefficient

The required gross airfoil maximum lift coefficient is estimated as:

$$C_{l_{\max\text{-}Gross}} = \frac{C_{L_{\max -w}}}{0.9} \tag{2.14}$$

$$C_{l_{\max\text{-}Gross}} = \frac{1.515}{0.9} = 1.683 \tag{2.15}$$

Therefore, the selected airfoil should be able to provide a maximum lift coefficient close to or higher than 1.683 when high-lift effects are considered.

Calculated Quantity	Value
Aircraft cruise lift coefficient, C_{L_C}	0.394
Wing cruise lift coefficient, $C_{L_{C_w}}$	0.415
Airfoil ideal lift coefficient, C_{l_i}	0.461
Aircraft maximum lift coefficient, $C_{L_{\max}}$	1.44
Wing maximum lift coefficient, $C_{L_{\max -w}}$	1.515
Required gross airfoil maximum lift coefficient, $C_{l_{\max\text{-}Gross}}$	1.683

Table 2.2: Summary of required lift coefficient calculations

2.3 Initial Research-Based Airfoil Selection

Based on the calculated requirements, several airfoils were considered for the wing design. The main comparison criteria were maximum lift coefficient, drag coefficient, pitching moment, stall behavior, cruise efficiency, and structural thickness.

Airfoil	t/c (%)	$C_{L\alpha}$ (1/rad)	α_0 (°)	$C_{L_{\max}}$	C_{m_0}	$C_{D_{\min}}$
NACA 2412	12	6.3	−2.0	1.52	−0.068	0.0065
NACA 4412	12	6.3	−4.0	1.80	−0.10	0.0068
NACA 23012	12	6.4	−1.2	1.52	−0.050	0.0080
NACA 23015	15	6.4	−1.0	1.52	−0.050	0.0080
NACA 63_1-412	12	6.2	−1.5	1.50	−0.080	0.0060

Table 2.3: Initial candidate airfoil comparison

From the initial research comparison, NACA 2412 was selected as a suitable airfoil because it provides a balanced combination of lift, drag, pitching moment, and safe stall behavior. It also has a common application in general aviation aircraft, which makes it a reasonable initial choice for the aerodynamic wing design.

2.4 Initial Selection Matrix

A weighted selection matrix was used to compare the candidate airfoils. The highest weight was given to stall quality and pitching moment because utility-category aircraft must have safe handling characteristics and acceptable trim behavior.

Airfoil	$C_{L_{\max}}/6$	$(C_L/C_D)_{\max}$	$C_{m_0}/8$	$C_{D_{\min}}/5$	Stall Quality/10	Total/36	%
Perfect	6	7	8	5	10	36	100%
NACA 2412	**6**	**6**	**7**	**5**	**9**	**33**	**92%**
NACA 4412	6	5	3	4	8	26	72%
NACA 23012	6	6	8	3	9	32	89%
NACA 23015	6	4	8	3	10	31	86%
NACA 63_1-412	5	6	3	5	6	25	69%

Table 2.4: Initial airfoil selection matrix

According to this matrix, NACA 2412 achieved the highest score of 92%. For this reason, it was first chosen for the wing aerodynamic design and the first CAD modelling trial.

2.5 NACA 2412 Design Calculations

For the NACA 2412 airfoil:

Property	Value
Maximum camber	2%
Camber position	40% chord
Maximum thickness	12% chord
Lift curve slope, C_{L_α}	6.3 rad^{-1}
Zero-lift angle, α_0	$-2°$
Maximum lift coefficient, $C_{L_{\max}}$	1.52
Minimum drag coefficient, $C_{D_{\min}}$	0.0065

Table 2.5: NACA 2412 properties used in the initial design

The required ideal airfoil lift coefficient was calculated as:

$$C_{l_i} = 0.461 \tag{2.16}$$

The angle of attack required to reach this lift coefficient can be estimated from the linear lift curve relation:

$$C_l = C_{L_\alpha}(\alpha - \alpha_0) \tag{2.17}$$

Therefore:

$$\alpha = \alpha_0 + \frac{C_l}{C_{L_\alpha}} \tag{2.18}$$

Substituting the NACA 2412 values:

$$\alpha = -2^\circ + \frac{0.461}{6.3} \times \frac{180^\circ}{\pi} \tag{2.19}$$

$$\alpha = -2^\circ + 4.19^\circ = 2.19^\circ \tag{2.20}$$

Thus, the wing setting angle was selected as:

$$i^T \approx 2^\circ \tag{2.21}$$

This means that the NACA 2412 wing can generate the required cruise lift coefficient while keeping the aircraft fuselage nearly level during cruise.

2.6 XFLR5 Analysis of NACA 23015

After completing the initial research-based selection, XFLR5 was used to verify the aerodynamic performance of the candidate airfoils. The analysis showed that NACA 23015 gives good aerodynamic behavior and provides the additional structural advantage of 15% thickness. This made it more suitable for the detailed CAD wing structure.

The XFLR5 analysis setup for NACA 23015 is summarized in Table 2.6.

XFLR5 Input / Output Parameter	Value
Airfoil	NACA 23015
Thickness	15.00%
Maximum thickness position	29.73% chord
Maximum camber	1.81%
Maximum camber position	15.32% chord
Number of panels	199
Reynolds number	4.55×10^6
Mach number	0.2866
Angle of attack used in the shown result	6°
N_{crit}	9.0
Forced upper transition	1.00
Forced lower transition	1.00
Lift coefficient, C_l	0.855
Drag coefficient, C_d	0.0069 approximately
Moment coefficient, C_m	−0.008 approximately
Lift-to-drag ratio, L/D	123.62

Table 2.6: XFLR5 setup and extracted results for NACA 23015

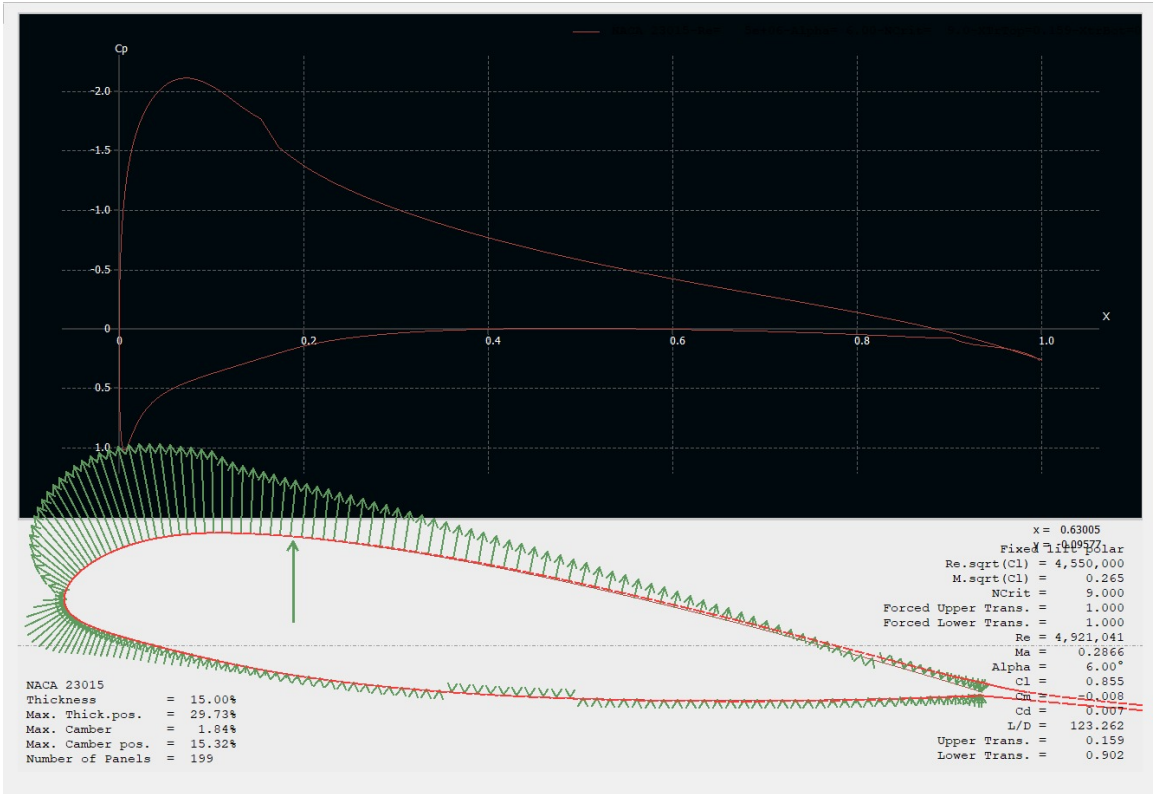


Figure 2.1: XFLR5 pressure coefficient distribution and boundary-layer visualization for NACA 23015 at $\alpha = 6°$

The pressure coefficient plot shows that the upper surface has strong suction near the

leading edge, which contributes to lift generation. The pressure recovery toward the trailing edge appears smooth, indicating acceptable aerodynamic behavior at the selected operating condition. The obtained lift coefficient of approximately $C_l = 0.855$ is higher than the required ideal airfoil lift coefficient of $C_{l_i} = 0.461$, meaning that the airfoil has enough lift capability at the analyzed condition.

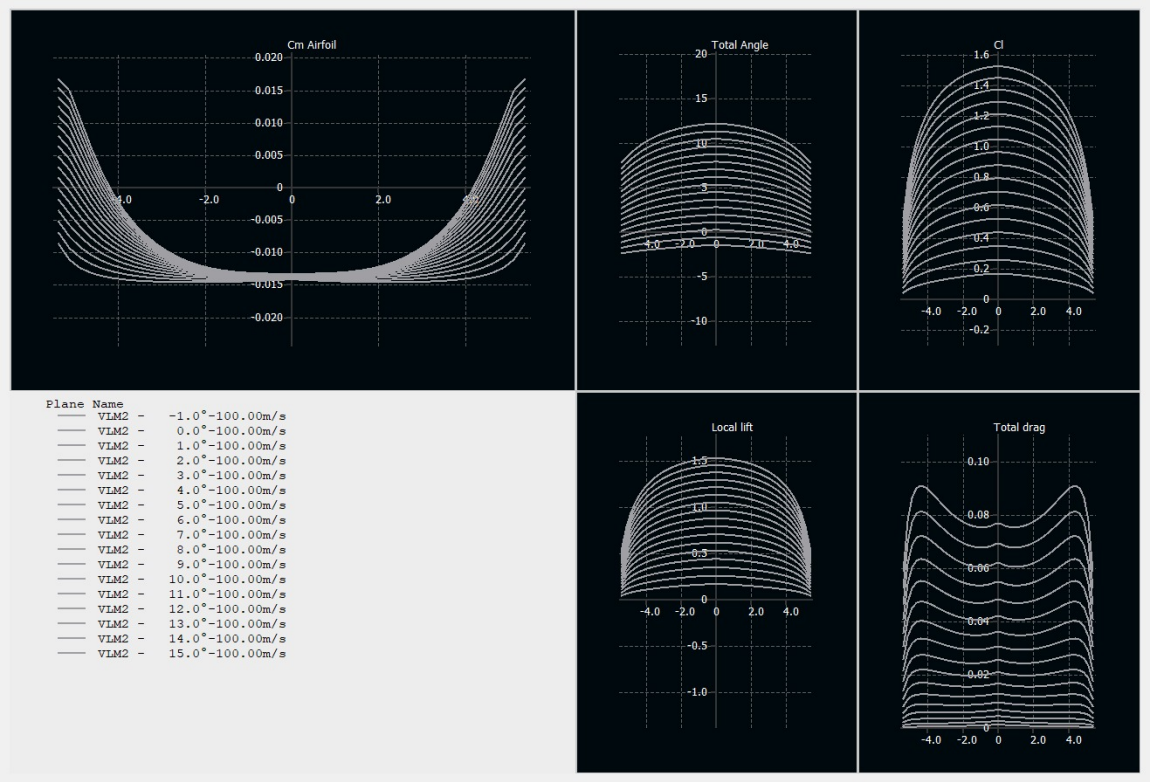


Figure 2.2: XFLR5 aerodynamic distributions for NACA 23015 showing moment coefficient, total angle, lift coefficient, local lift, and total drag behavior

Figure 2.2 shows the variation of aerodynamic coefficients along the analyzed operating range. The lift distribution increases smoothly with angle of attack, while the drag remains relatively low at moderate angles. This supports the use of NACA 23015 as a strong candidate for the mechanical wing model.

2.7 XFLR5 Comparison Between Candidate Airfoils

The candidate airfoils were compared in XFLR5 under take-off and cruise operating conditions. The comparison included the following airfoils:

- NACA 23012
- NACA 23015
- NACA 2412
- NACA 4412
- NACA 63-412

The main comparison plots include C_l versus C_d, C_l versus α, C_d versus α, C_l/C_d versus α, and C_m versus α.

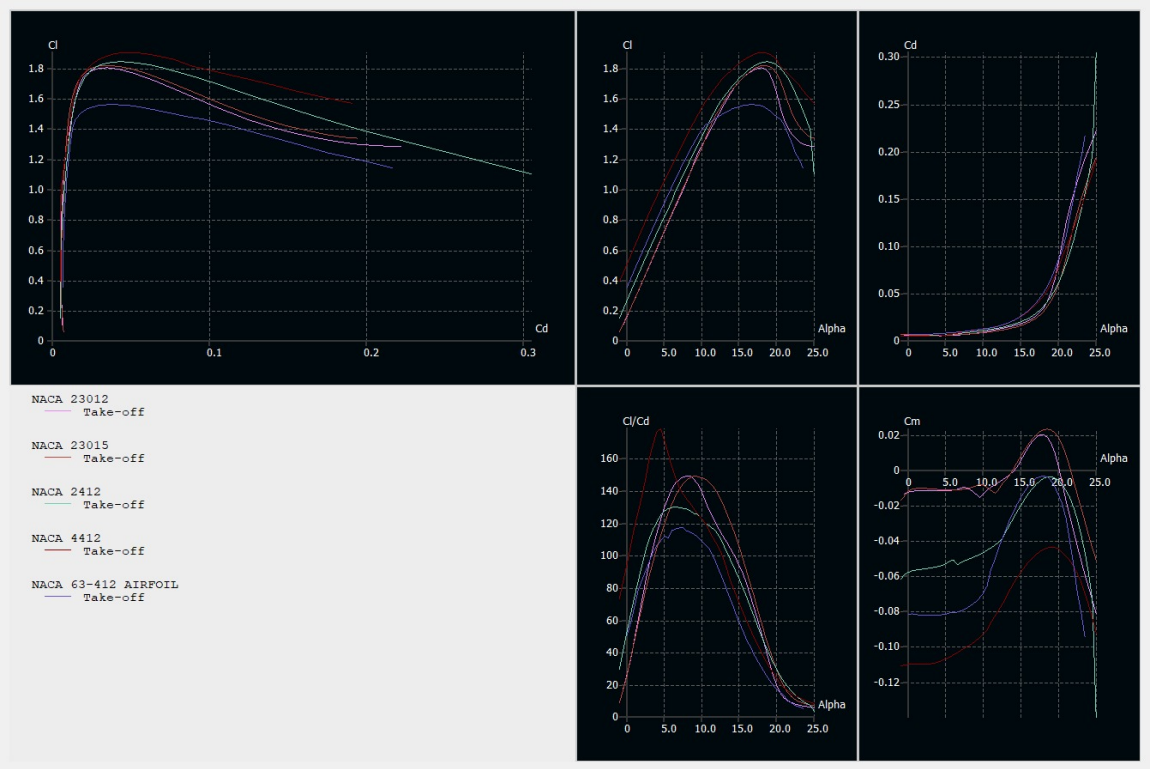


Figure 2.3: XFLR5 take-off comparison between candidate airfoils

At take-off condition, the XFLR5 comparison shows that NACA 23015 produces high lift and maintains good aerodynamic efficiency. It also has a favorable C_l/C_d range compared with the other tested airfoils. This is important because the take-off phase requires high lift generation at relatively low speed.

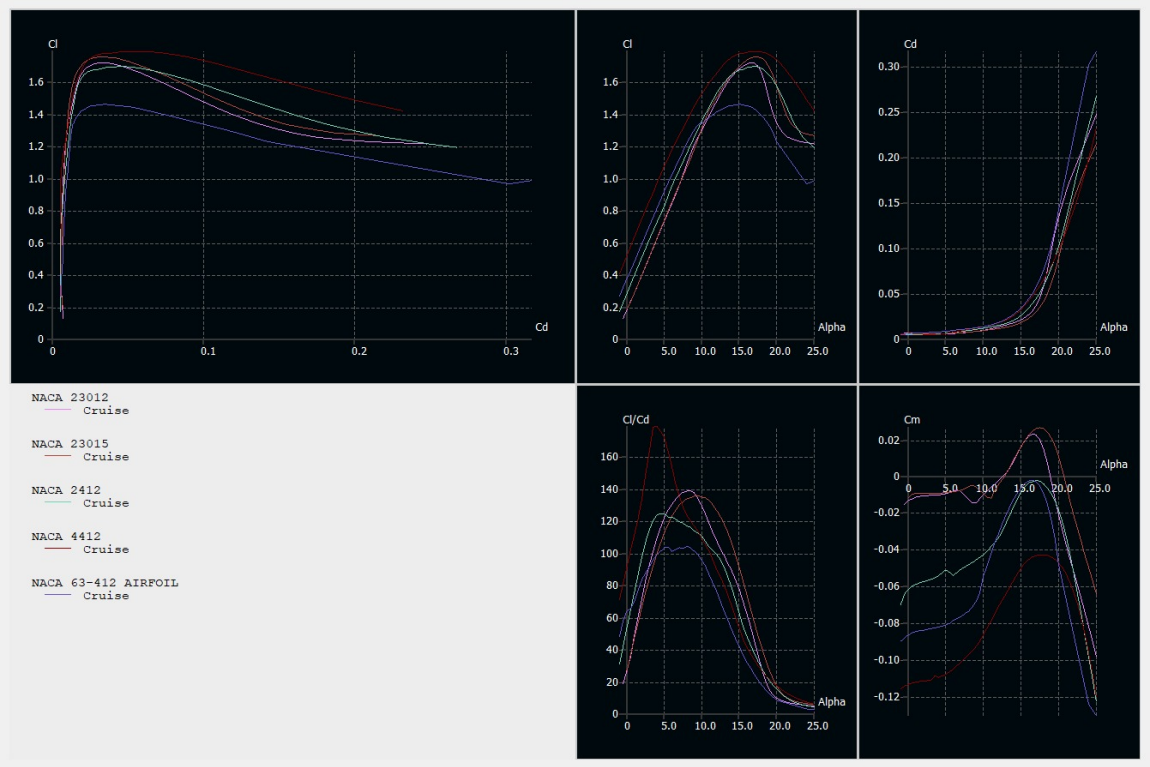


Figure 2.4: XFLR5 cruise comparison between candidate airfoils

At cruise condition, NACA 23015 also shows good lift-to-drag behavior. The comparison indicates that it can provide sufficient lift while keeping drag reasonably low. The thicker 15% section also gives an important structural benefit by increasing the available internal depth for spars and ribs.

2.8 Comparison Between NACA 2412 and NACA 23015

The initial analytical and research-based selection favored NACA 2412 because of its balanced aerodynamic properties and known suitability for general aviation aircraft. However, the XFLR5 results showed that NACA 23015 is also a strong candidate and may be more suitable for the detailed mechanical CAD design because of its larger thickness.

Comparison Point	NACA 2412	NACA 23015
Thickness ratio	12%	15%
Maximum camber	2%	1.81%
Initial selection matrix score	92%	86%
Research-based suitability	Very good	Good
XFLR5 lift behavior	Good	Very good
Internal structural space	Moderate	Higher
CAD structural suitability	Acceptable	Better
Main advantage	Balanced GA performance	Better internal depth and strong XFLR5 performance

Table 2.7: Comparison between NACA 2412 and NACA 23015

From this comparison, NACA 2412 was suitable for the initial aerodynamic design, while NACA 23015 was more suitable for the mechanical CAD phase. The reason is that NACA 23015 provides more thickness, allowing better placement of the I-beam spars, C-shaped stringers, ribs, and skin inside the wing. In addition, the XFLR5 results confirmed that NACA 23015 can achieve good lift and lift-to-drag performance.

2.9 Final Airfoil Decision for CAD Modelling

Based on the complete study, the project used two levels of airfoil decision-making:

- **NACA 2412** was selected from the initial research-based airfoil selection because it achieved the highest score in the selection matrix and showed balanced aerodynamic behavior.
- **NACA 23015** was selected for the detailed CAD wing design because XFLR5 showed good aerodynamic performance, and the 15% thickness provided better internal structural space.

Therefore, the final mechanical CAD wing design was developed using NACA 23015, while the NACA 2412 calculations were kept as the initial aerodynamic selection reference.

Chapter 3

Wing Geometric Design

3.1 Wing Planform Selection

A straight tapered trapezoidal wing was selected. Since the aircraft cruises at approximately $M = 0.36$, wing sweep is not required. An unswept wing improves structural efficiency, simplifies manufacturing, and is suitable for subsonic utility general aviation aircraft.

3.2 Wing Setting Angle

The wing setting angle is selected using the angle corresponding to the ideal airfoil lift coefficient.

$$\alpha = \alpha_0 + \frac{C_{l_i}}{C_{L_\alpha}} = -2^\circ + \frac{0.461}{6.3} \times \frac{180^\circ}{\pi} = 2.19^\circ \approx 2^\circ \tag{3.1}$$

Therefore, the selected wing setting angle is:

$$i^T = 2^\circ \tag{3.2}$$

3.3 Aspect Ratio and Wingspan

The aspect ratio is selected as $AR = 7.0$, which is suitable for general aviation aircraft.

$$b = \sqrt{AR \times S} = \sqrt{7.0 \times 17.1} = 10.94 \text{ m} \tag{3.3}$$

3.4 Taper Ratio and Chord Dimensions

The taper ratio is selected as $\lambda = 0.60$.

$$\overline{C} = \frac{S}{b} = \frac{17.1}{10.94} = 1.564 \text{ m} \tag{3.4}$$

$$C_r = \frac{1.5(1+\lambda)\overline{C}}{1+\lambda+\lambda^2} = \frac{1.5(1.60)(1.564)}{1+0.60+0.36} = 1.916 \text{ m} \tag{3.5}$$

$$C_t = \lambda C_r = 0.60 \times 1.916 = 1.150 \text{ m} \tag{3.6}$$

3.5 Dihedral Angle

A dihedral angle of 5° is selected to provide positive lateral stability. This value is typical for utility-category general aviation aircraft.

3.6 Wing Twist

A geometric washout of $-1°$ is applied from root to tip. This helps the wing root stall before the tip, preserving aileron effectiveness near stall.

3.7 Wing Geometry Summary

Parameter	Selected Value	Justification
Wing area S	17.1 m^2	Given specification
Aspect ratio AR	7.0	Suitable for GA aircraft
Wingspan b	10.94 m	Derived from $b = \sqrt{AR \times S}$
Taper ratio λ	0.60	Near-elliptic lift distribution
Root chord C_r	1.916 m	Derived from geometry
Tip chord C_t	1.150 m	Derived from $C_t = \lambda C_r$
Mean aerodynamic chord $\overline{C}$	1.564 m	Derived from S/b
Sweep angle Λ	0°	Low subsonic cruise speed
Setting angle i^T	2°	Based on ideal lift coefficient
Dihedral angle Γ	5°	Lateral stability
Geometric twist	$-1°$	Tip-stall prevention
Airfoil section	NACA 2412	Highest selection matrix score

Table 3.1: Final wing geometry parameters

Chapter 4

Mechanical Wing Design Using NACA 23015

4.1 Design Overview

This chapter presents the mechanical and CAD-based wing design developed using the NACA 23015 airfoil. Although the previous aerodynamic selection favored NACA 2412, the NACA 23015 airfoil was used in this mechanical design phase because its 15% thickness provides a larger internal depth. This additional thickness gives more available space for the main spar, rear spar, C-shaped stringers, ribs, and skin.

The wing structure was modeled as a semi-span tapered wing. The internal layout consists of two I-beam spars, C-shaped stringers, multiple ribs distributed along the span, and an external skin. The design was prepared to represent a preliminary CAD model suitable for further stress analysis, weight estimation, and manufacturing refinement.

The main structural components included in the CAD model are:

- Main I-beam spar
- Rear I-beam spar
- C-shaped stringers
- Fifteen ribs along the semi-span
- Wing skin
- Full internal and external CAD assembly

4.2 Main I-Beam Spar Design

The main spar is the primary load-carrying component of the wing. It resists most of the bending moment generated by the aerodynamic lift force. An I-beam cross-section was selected because the upper and lower caps carry the bending stresses, while the vertical web carries the shear force.

Since the bending moment is maximum near the wing root and decreases toward the tip, the main spar dimensions are larger at the root and reduced at the tip.

4.2.1 Main Spar at the Root

Parameter	Value
Web thickness	18 mm
Web length	188 mm
Cap length	80 mm
Cap thickness	18 mm
Material	Aluminium alloy 2024-T3

Table 4.1: Main I-beam spar characteristics at the wing root

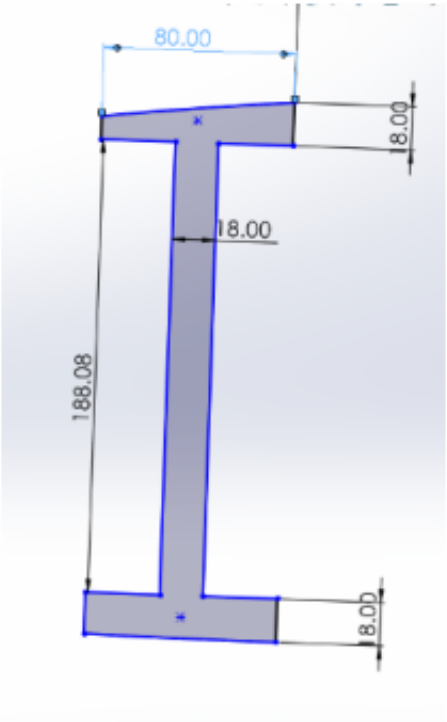


Figure 4.1: Main I-beam spar cross-section at the wing root

4.2.2 Main Spar at the Tip

Parameter	Value
Web thickness	10.8 mm
Web length	112 mm
Cap length	48 mm
Cap thickness	10.8 mm
Material	Aluminium alloy 2024-T3

Table 4.2: Main I-beam spar characteristics at the wing tip

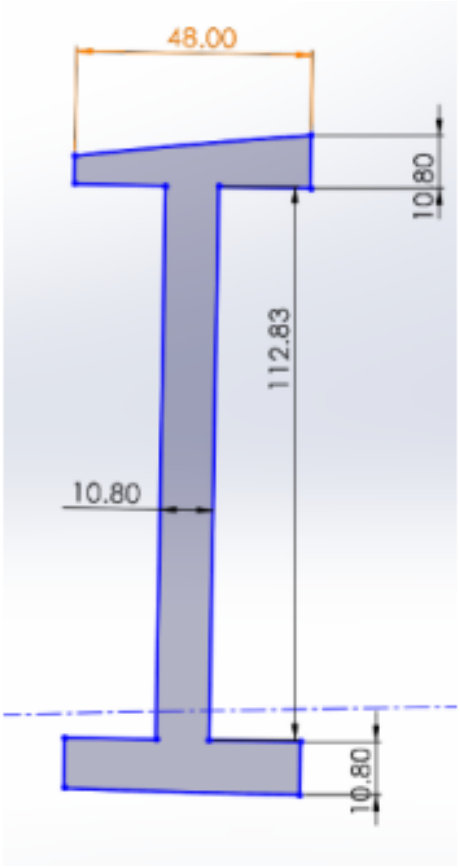


Figure 4.2: Main I-beam spar cross-section at the wing tip

4.3 Rear I-Beam Spar Design

The rear spar supports the aft region of the wing box and improves the torsional stiffness of the wing. It also helps define the structural boundary for control surfaces such as the aileron and flap. Similar to the main spar, the rear spar dimensions are larger at the root and smaller at the tip because the structural loads decrease along the span.

4.3.1 Rear Spar at the Root

Parameter	Value
Web thickness	18 mm
Web length	128 mm
Cap length	80 mm
Cap thickness	18 mm
Material	Aluminium alloy 2024-T3

Table 4.3: Rear I-beam spar characteristics at the wing root

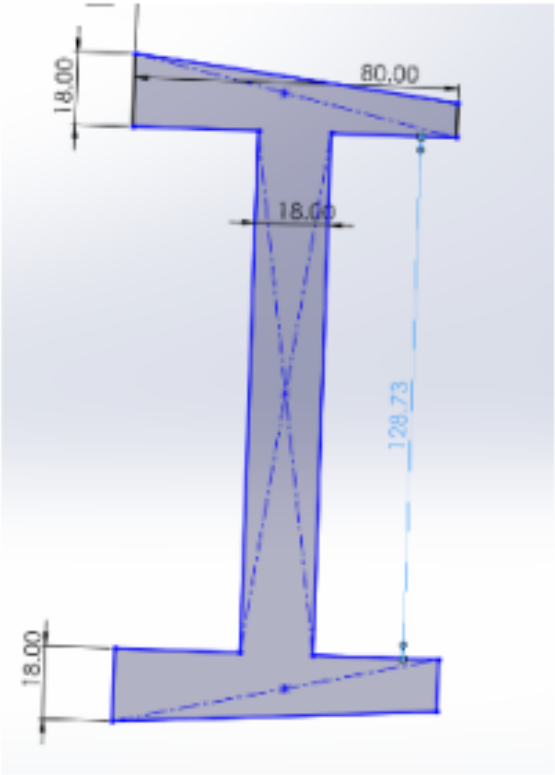


Figure 4.3: Rear I-beam spar cross-section at the wing root

4.3.2 Rear Spar at the Tip

Parameter	Value
Web thickness	10.8 mm
Web length	77.23 mm
Cap length	48 mm
Cap thickness	10.8 mm
Material	Aluminium alloy 2024-T3

Table 4.4: Rear I-beam spar characteristics at the wing tip

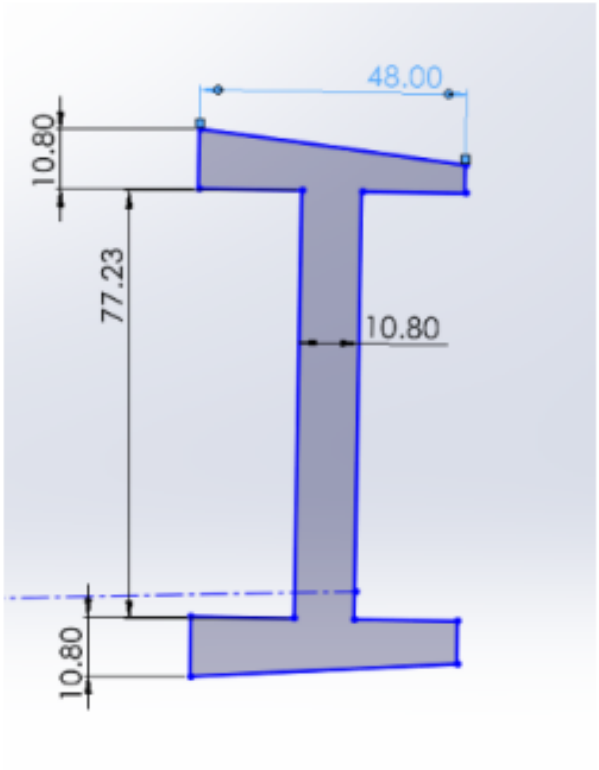


Figure 4.4: Rear I-beam spar cross-section at the wing tip

4.4 C-Shaped Stringer Design

Stringers are longitudinal stiffeners used to support the wing skin and reduce local skin buckling. They also help carry part of the axial stresses caused by wing bending. A C-shaped stringer was selected because it is simple to manufacture, easy to attach to ribs and skin, and suitable for thin-walled aircraft structures.

4.4.1 C-Stringers at the Root

Parameter	Value
Number of stringers	7
Distance between stringers	97.78 mm
Outer length	30 mm
Inner length	10 mm
Thickness	20 mm
Material	Aluminium alloy 2024-T3

Table 4.5: C-shaped stringer characteristics at the wing root

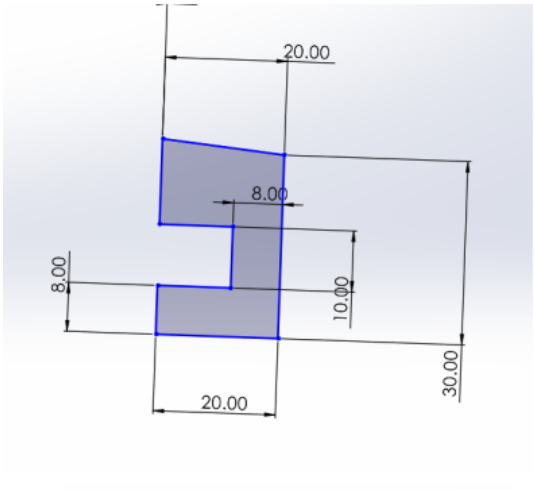


Figure 4.5: C-shaped stringer cross-section at the wing root

4.4.2 C-Stringers at the Tip

Parameter	Value
Number of stringers	7
Distance between stringers	58.67 mm
Outer length	18 mm
Inner length	6 mm
Thickness	12 mm
Material	Aluminium alloy 2024-T3

Table 4.6: C-shaped stringer characteristics at the wing tip

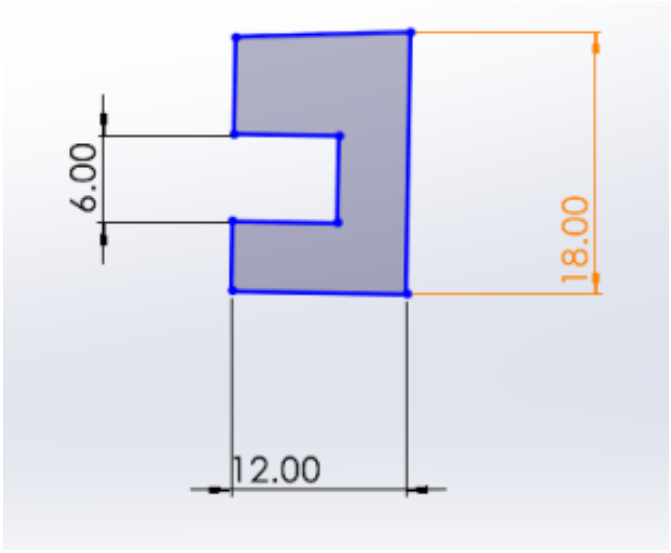


Figure 4.6: C-shaped stringer cross-section at the wing tip

The stringer dimensions decrease from root to tip because both the local chord length and the expected structural loading decrease along the span. Seven stringers were used to provide enough skin support while keeping the internal structure simple and manufacturable.

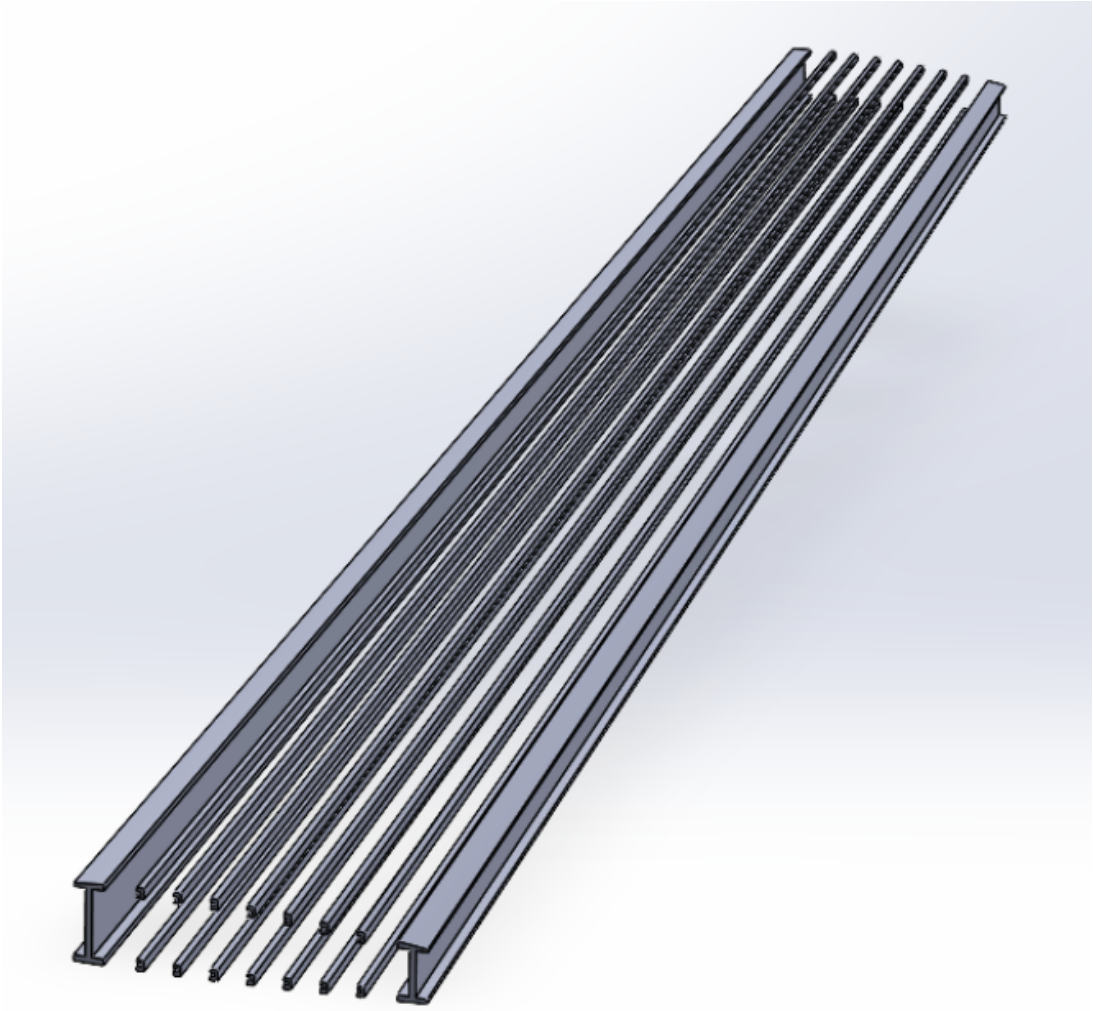


Figure 4.7: Isometric CAD view of the spars and C-shaped stringers

4.5 Rib Design and Chord Distribution

Ribs are used to maintain the NACA 23015 airfoil shape along the wing span. They transfer aerodynamic loads from the skin to the spars and preserve the external aerodynamic contour of the wing. In the CAD model, each rib was created as a scaled NACA 23015 profile based on its local chord length.

The local chord was calculated using the linear taper relation:

$$c(y) = C_r - (C_r - C_t)\,\frac{y}{b/2} \tag{4.1}$$

where $C_r = 1.916$ m is the root chord, $C_t = 1.150$ m is the tip chord, and $b/2 = 5.47$ m is the semi-span.

Rib No.	y from Root	Local Chord
R1	0.000 m	1.916 m
R2	0.391 m	1.861 m
R3	0.781 m	1.807 m
R4	1.172 m	1.752 m
R5	1.563 m	1.697 m
R6	1.954 m	1.642 m
R7	2.344 m	1.588 m
R8	2.735 m	1.533 m
R9	3.126 m	1.478 m
R10	3.516 m	1.424 m
R11	3.907 m	1.369 m
R12	4.298 m	1.314 m
R13	4.689 m	1.260 m
R14	5.079 m	1.205 m
R15	5.470 m	1.150 m

Table 4.7: Rib station locations and local chord lengths for the NACA 23015 CAD model

Therefore, in the CAD model, every rib should be a scaled NACA 23015 section using its corresponding local chord value.

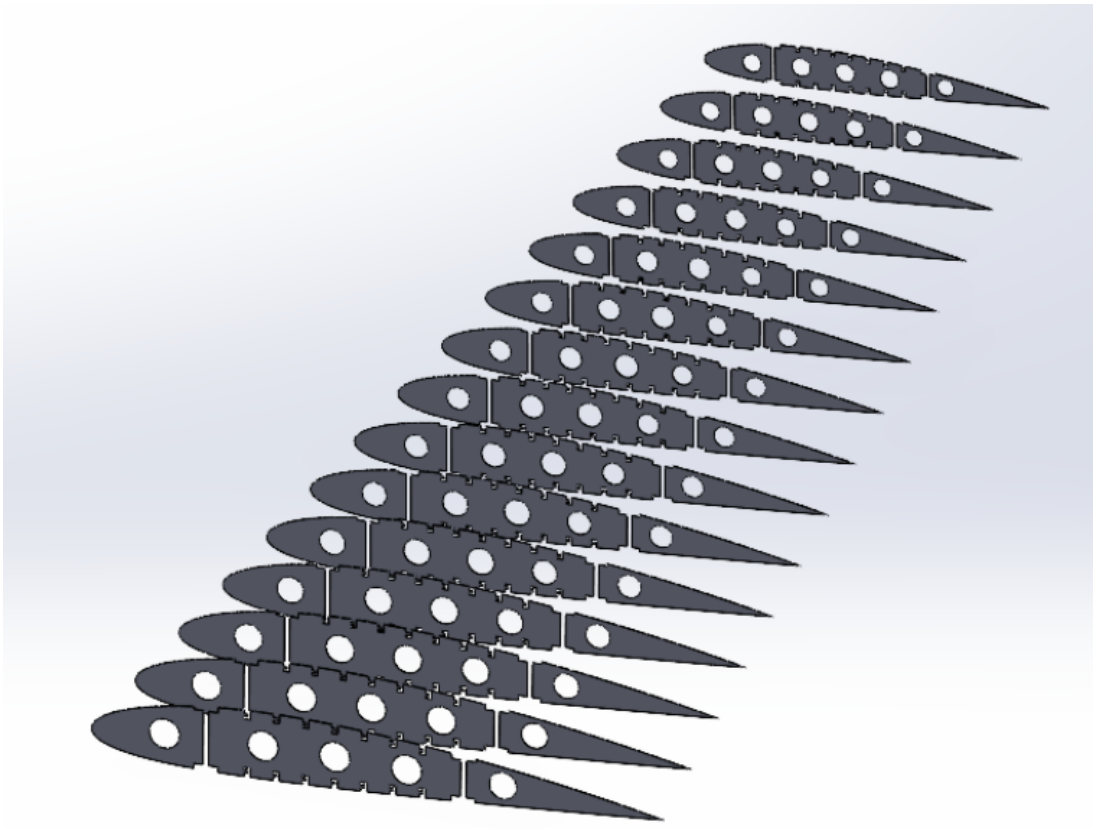


Figure 4.8: Isometric CAD view of the ribs and their spanwise positions

4.6 Wing Skin Design

The wing skin forms the outer aerodynamic surface of the wing. It carries distributed pressure loads and transfers them to the ribs and stringers. The skin also contributes to the torsional stiffness of the wing box. In this CAD model, a skin thickness of 10 mm was used.

Parameter	Value
Skin thickness	10 mm
Airfoil profile	NACA 23015
Material	Aluminium alloy 2024-T3

Table 4.8: Wing skin design characteristics

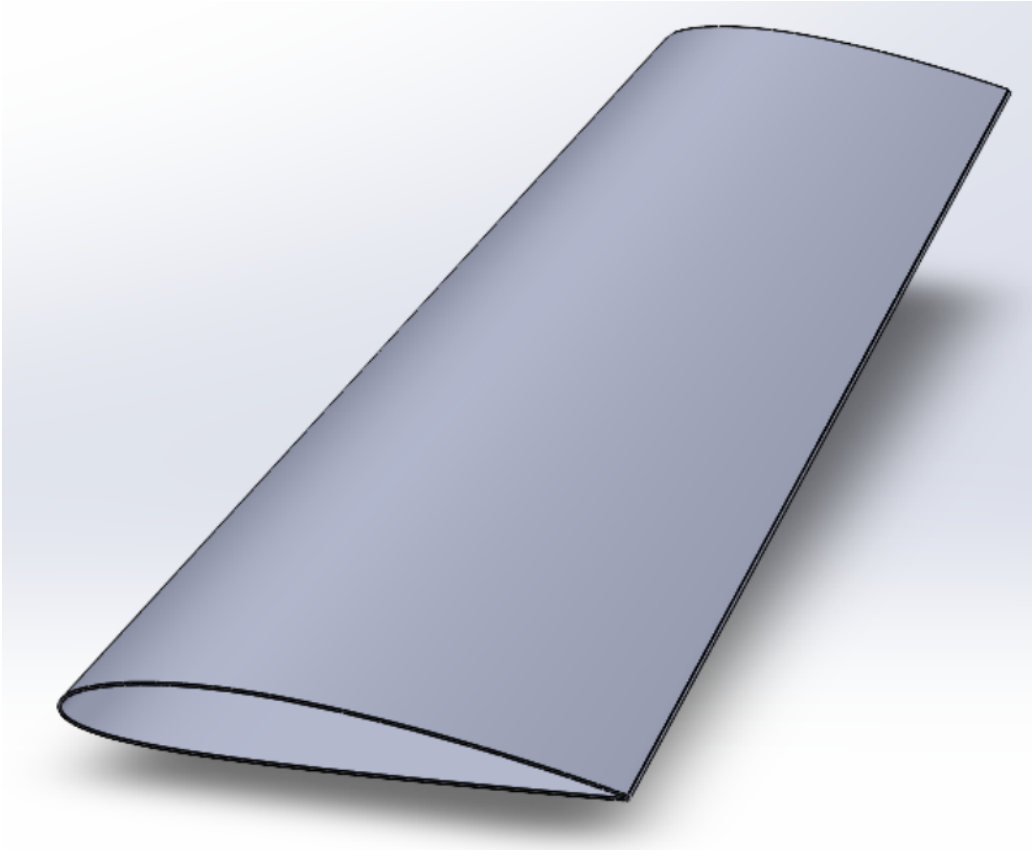


Figure 4.9: Isometric CAD view of the wing skin

4.7 Complete CAD Assembly

The complete CAD assembly combines the NACA 23015 ribs, main spar, rear spar, C-shaped stringers, and external skin. This model represents the complete preliminary mechanical design of the wing. It can be used as the basis for future structural analysis, finite element simulation, mass estimation, and manufacturing improvements.

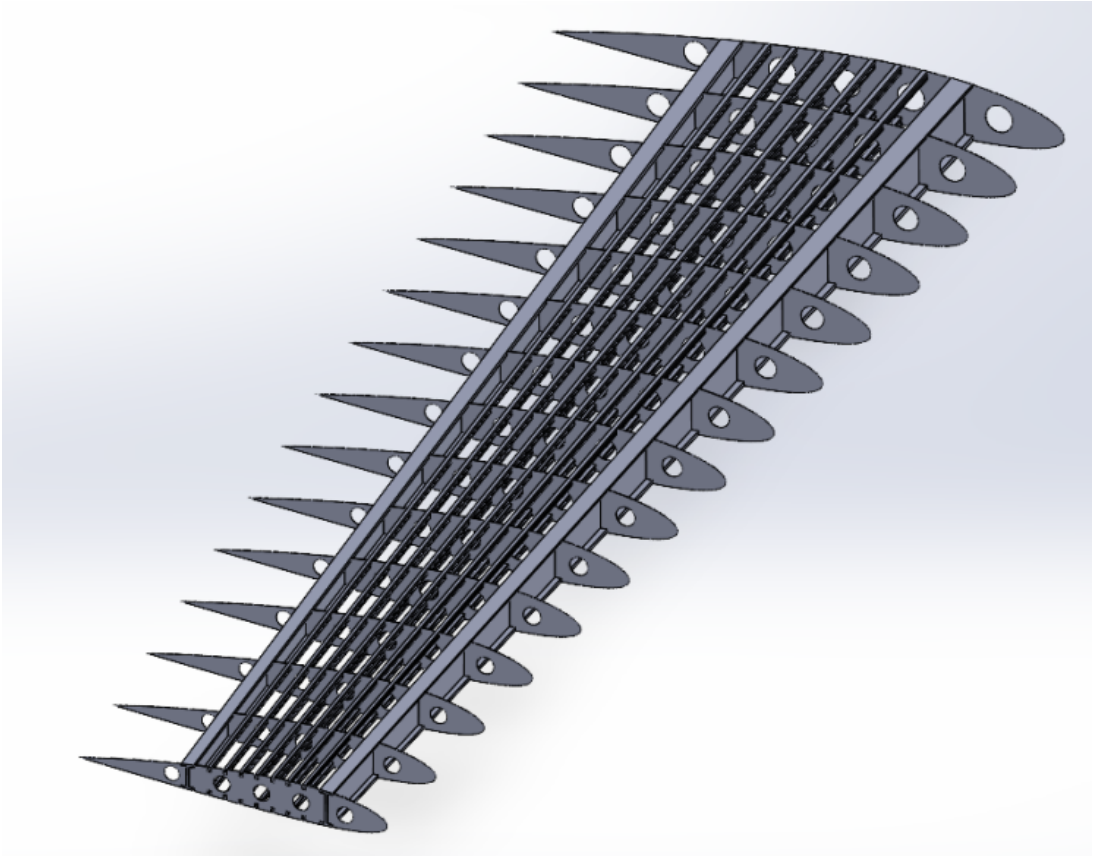


Figure 4.10: Full internal CAD design showing spars, stringers, ribs, and skin

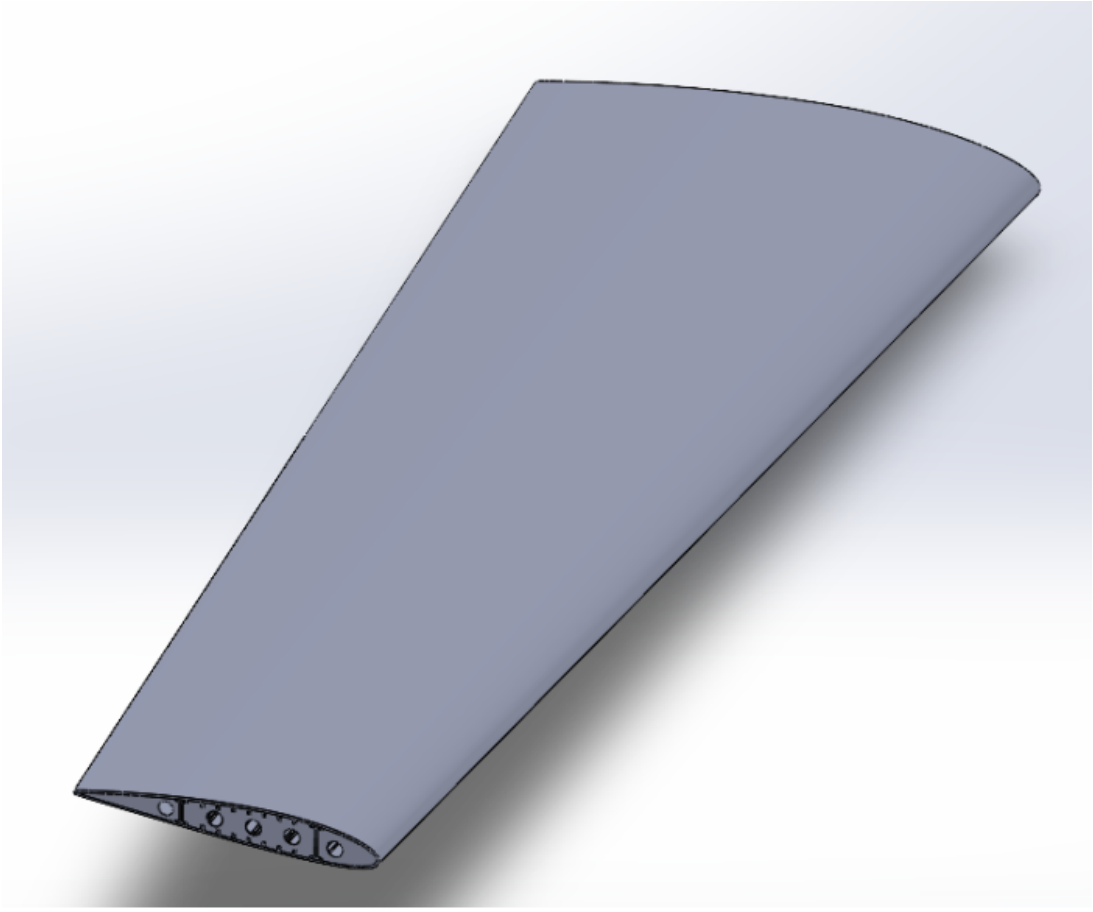


Figure 4.11: Complete isometric CAD view of the designed NACA 23015 wing

4.8 Mechanical Design Summary

The NACA 23015 airfoil provided enough internal depth for arranging the structural components inside the wing. The main I-beam spar was selected as the primary bending load-carrying member, while the rear spar was used to support the aft wing box and improve torsional rigidity. C-shaped stringers were added to support the skin and reduce local buckling. Fifteen ribs were distributed along the semi-span, and each rib was scaled according to its local chord.

Overall, the CAD design provides a complete preliminary mechanical wing structure consisting of spars, stringers, ribs, and skin. This design is suitable for the next phase of the project, where stress analysis, deflection analysis, and weight optimization can be performed.

Chapter 5

Spanwise Lift Distribution

5.1 Theoretical Background

The spanwise lift distribution was calculated using Prandtl's Lifting-Line Theory. This method models the finite wing using a bound vortex and a trailing vortex sheet. It is widely used for preliminary finite-wing aerodynamic analysis.

$$\mu(\alpha_0 - \alpha) = \sum_{n=1}^{N} A_n \sin(n\theta) \left(1 + \frac{\mu n}{\sin\theta}\right) \tag{5.1}$$

$$\mu = \frac{\overline{C} C_{L_\alpha}}{4b} \tag{5.2}$$

$$C_L(y) = \frac{4b}{\overline{C}} \sum_{n=1}^{N} A_n \sin(n\theta) \tag{5.3}$$

$$C_{L_w} = \pi A R A_1 \tag{5.4}$$

5.2 MATLAB Implementation

The lifting-line solver was implemented in MATLAB. The semi-span was divided into nine angular stations, and the resulting linear system was solved to obtain the Fourier coefficients.

```
clc
clear
N = 9;
S = 17.1;
AR = 7;
lambda = 0.60;
alpha_twist = -1;
i_w = 2;
a_2d = 6.3;
alpha_0 = -2;
b = sqrt(AR*S);
MAC = S/b;
Croot = (1.5*(1+lambda)*MAC)/(1+lambda+lambda^2);
theta = pi/(2*N):pi/(2*N):pi/2;
alpha = i_w+alpha_twist:-alpha_twist/(N-1):i_w;
z = (b/2)*cos(theta);
c = Croot*(1-(1-lambda)*cos(theta));
mu = c * a_2d / (4 * b);
LHS = mu .* (alpha-alpha_0)/57.3;

for i=1:N
    for j=1:N
        B(i,j) = sin((2*j-1)*theta(i))*(1+(mu(i)*(2*j-1))/sin(theta(i
            )));
    end
end

A = B\transpose(LHS);

for i = 1:N
    sum1(i) = 0;
    sum2(i) = 0;
    for j = 1:N
        sum1(i) = sum1(i) + (2*j-1)*A(j)*sin((2*j-1)*theta(i));
        sum2(i) = sum2(i) + A(j)*sin((2*j-1)*theta(i));
    end
end

CL = 4*b*sum2 ./ c;
CL_wing = pi * AR * A(1)
```

Figure 5.1: MATLAB implementation of the lifting-line solver

5.3 Solver Inputs

Input Parameter	Value	Source
Wing area S	17.1 m^2	Given
Aspect ratio AR	7.0	Chapter 3
Taper ratio λ	0.60	Chapter 3
Lift curve slope C_{L_α}	6.3 /rad	NACA 2412
Zero-lift angle α_0	$-2°$	NACA 2412
Wing setting angle i^T	$2°$	Chapter 3
Geometric twist	$-1°$	Chapter 3
Number of segments N	9	MATLAB solver setup

Table 5.1: Inputs used in the MATLAB lifting-line solver

5.4 Spanwise Lift Distribution Results

Seg.	y (m)	θ (°)	Chord c (m)	$C_L(y)$	Note
1	5.37	10.0	1.199	0.291	Near tip
2	5.15	20.0	1.271	0.302	–
3	4.77	30.0	1.385	0.308	–
4	4.26	40.0	1.527	0.305	–
5	3.63	50.0	1.677	0.296	Mid-span
6	2.87	60.0	1.820	0.286	–
7	2.01	70.0	1.936	0.277	–
8	1.08	80.0	1.986	0.270	–
9	0.19	90.0	1.916	0.266	Root

Table 5.2: Spanwise lift coefficient distribution from MATLAB

5.5 Total Wing Lift Coefficient Verification

The total wing lift coefficient obtained from the first Fourier coefficient is:

$$C_{L_w} = \pi AR A_1 \approx 0.395 \tag{5.5}$$

The required cruise lift coefficient is:

$$C_{L,required} = 0.394 \tag{5.6}$$

The small difference between the required value and the MATLAB result confirms that the selected wing geometry is suitable for the cruise condition.

5.6 Estimated Wing Lift at Take-Off

At take-off speed:

$$V_{TA} = 1.3V_s = 1.3 \times 51.44 = 66.87 \text{ m/s} \tag{5.7}$$

The estimated lift is:

$$L = \frac{1}{2}\rho V_{TA}^2 S C_{L_w} = \frac{1}{2}(1.225)(66.87)^2(17.1)(0.395) = 29{,}432 \text{ N} \tag{5.8}$$

Since the aircraft weight is:

$$W = 2717 \times 9.81 = 26{,}664 \text{ N} \tag{5.9}$$

then:

$$\frac{L}{W} = 1.10 \tag{5.10}$$

This means the wing provides approximately 10% lift margin at take-off.

Chapter 6

Conclusion

The preliminary wing design satisfies the main aerodynamic requirements for the utility-category general aviation aircraft. NACA 2412 was selected as the airfoil because it provides a balanced combination of maximum lift, cruise efficiency, acceptable pitching moment, docile stall behavior, and sufficient structural thickness. The final wing geometry includes an aspect ratio of 7.0, a taper ratio of 0.60, a span of 10.94 m, a root chord of 1.916 m, a tip chord of 1.150 m, a setting angle of 2°, a dihedral angle of 5°, and a washout angle of $-1°$.

The MATLAB lifting-line analysis produced a total wing lift coefficient of approximately 0.395, which closely matches the required cruise lift coefficient of 0.394. The take-off lift estimation also showed that the wing can generate lift greater than the aircraft weight, with a lift margin of approximately 10%. Therefore, the selected wing configuration is suitable for further structural design and CAD development.

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